

Computational Phytopharmacology for premenopausal Endocrine Support: A Machine Learning Perspective

*Dissertation submitted to the VIT in partial fulfillment of the
requirements for the degree of*

**MASTER OF SCIENCE
IN
BIOTECHNOLOGY (5 YEAR INTEGRATED)**

By

NIRANJANI M

(Reg. No. 22MSI0154)



VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

**Department of Integrative Biology
School of Bio Sciences and Technology
Vellore Institute of Technology, Vellore- 632014**

April, 2026

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DECLARATION

I, hereby, declare that the thesis entitled, “**Computational Phytopharmacology for premenopausal Endocrine Support: A Machine Learning Perspective**” submitted by **NIRANJANI M, Reg. No. 22MSI0154** to VIT, Vellore in fulfillment of the requirement for the award of the degree of *M. Sc., [5year Integrated] Biotechnology* a record of bonafide research work carried out by me under the guidance of (**SAJITHA LULU S Associate Professor Sr.**) during the course of the year 2025 – 2026. I further declare that the work reported in this dissertation has not been submitted, and will not be submitted, either in part or in full, for the award of any other Degree or Diploma of this University or of any other Institute or University.

Date:

Signature of the Candidate

CERTIFICATE OF THE EXTERNAL GUIDE

This is to certify that the dissertation entitled, “**Computational Phytopharmacology for premenopausal Endocrine Support: A Machine Learning Perspective,**” is submitted by Ms **NIRANJANI M** Reg. No. **22MSI0154** to the School of Bio Sciences and Technology, VIT, Vellore - 632 014, Tamil Nadu in partial fulfillment of the requirements for the Degree of M.Sc. Biotechnology is a record of work carried by her/him under my supervision (Name of external guide, Designation, Institution). The contents of this dissertation, in full or in parts, have not been submitted to any other Institute or University for the award of any Degree or Diploma.

Signature of the Guide

CERTIFICATE OF THE INTERNAL GUIDE

This is to certify that the dissertation entitled, “**Computational Phytopharmacology for premenopausal Endocrine Support: A Machine Learning Perspective**” is submitted by Ms. **NIRANJANI M** Reg. No. **22MSI0154** to the School of Bio Sciences and Technology, VIT, Vellore- 632 014, Tamil Nadu in partial fulfillment of the requirements for the Degree of M.Sc.[5year Integrated] Biotechnology is a record of work carried by her/him under my supervision (**SAJITHA LULU S Associate Professor Sr.**). The contents of this dissertation, in full or in parts, have not been submitted to any other Institute or University for the award of any Degree or Diploma.

Signature of the
Internal Guide

Head, Department of Integrative Biology

Dean SBST

Internal Examiner

External Examiner

ABSTRACT

The objective of this work is to predict bioactive phytochemicals which have the capability of endocrine modulation activities and prioritize them for premenopausal use through integrated computational phytopharmacology and machine learning techniques. The total number of 349 phytochemicals was gathered from literature and filtered based on structural validity, biological relevancy, toxicity, and pharmacokinetics, which led to the identification of 139 good quality phytochemicals. The biological relevance was validated by chemical-protein interaction study, and toxicity (ProTox-II) and ADME (SwissADME) were carried out to validate safety and druglikeness of the molecules. Clustering using machine learning approach through KNIME was done based on molecular descriptors to prioritize the phytochemicals. Network interaction analysis found that TP53, TNF, IL6, and CTNNB1 are the important regulatory genes. Molecular docking analysis identified the high binding affinity of the selected phytochemicals; specifically, Campesterol exhibited high binding affinity toward TNF with -12.3 kcal/mol affinity. Other molecules with high interaction include Ursolic acid, Flavone, and Artemisinin.

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Chapter 1

Introduction

One of the most important roles of the endocrine system is its involvement in managing various physiological functions such as metabolism, growth, reproduction, and homeostasis. It functions based on the interaction between various glands and hormones in order to control body activities and maintain physiological balance. The endocrine system consisting of glands like pituitary, hypothalamus, and adrenal glands manages various physiological functions; thus, any disorder of the endocrine system results in numerous physiological conditions, including diabetes mellitus, hormonal disturbances, and thyroid gland diseases (12). All these diseases result from disturbances in the process of hormone production and regulation caused by genetic, environmental, and lifestyle factors (12).

Women's health is especially affected by the impact of hormonal fluctuation throughout their lives. Women's health is considerably influenced by fluctuations of hormones during their lifetime, as hormones affect many physiological processes related to such biological systems as cardiovascular, reproductive, nervous, and immune, leading to conditions like obesity, cardiovascular diseases, immune disorders, osteoporosis, and even cancer (9). The hormonal equilibrium involving the endocrine system and autocrine system tends to be disrupted with increasing age, especially after 45 years, causing an imbalance between estrogen and progesterone, thus giving rise to various syndromes and diseases affecting different physiological functions (2). The impact of hormone imbalance can have a major influence on various tissues which require steroid hormones in high quantity during reproductive ages such as brain, endometrium, bones, and skin (2), which is a progressive process even after menopause.

Menopause is a crucial period where ovarian function declines and there is a considerable reduction in the production of estrogen, causing hot flushes, vaginal dryness, bone loss, insomnia, and mood alterations (10).

For females specifically, the premenopausal stage is when endocrine regulation plays an important role because changes in hormone levels can have implications on reproductive processes, emotional wellness, and metabolism. Changes in hormones like estrogen and progesterone result in issues like irregular menstruation, fatigue, emotional instabilities, and metabolic disturbances (13).

One factor that can contribute to the disruption of endocrine regulation is stress, which is linked to the impairment of the normal functioning of the hypothalamic-pituitary-adrenal (HPA) axis due to increased cortisol levels in the body. Stress affects endocrine regulation by increasing cortisol secretion, which disrupts hormonal signaling and inhibits the effects of reproductive hormones leading to endocrine dysfunction (13). Endocrine dysfunction is connected to mental disorders, such as depression, which has been linked to alterations in neurotransmission, neuroplasticity, and HPA-axis functions, and is common in females (19).

It shows the strong relationship between the two systems.

Environmental factors, in addition to the physiological ones, also have a major influence on the imbalance in the endocrine system. Endocrine-disrupting chemicals (EDC) are capable of interfering with hormone production, secretion, transportation, and interaction with their receptors, thus, having an adverse effect on reproduction and development (21). Most endocrine-disrupting chemicals have estrogenic activity because of their ability to interact with estrogen receptors and modify signaling pathways, which results in disruption of hormonal balance (21). Such disruptions are likely to affect individuals for a prolonged period, particularly when exposed during sensitive phases of development. Aging, metabolic disorders, and lifestyle choices are additional elements that contribute to hormonal imbalance, affecting reproductive functions and overall health status (16). In addition to affecting reproduction and metabolism, such hormonal imbalances are also linked to several pathological states. One of the most common benign tumors in females is uterine leiomyoma (fibroids), which demonstrate the close relationship between hormones, inflammation, and cellular proliferation (4). Uterine fibroids are predominantly under the control of estrogen and progesterone hormones and are associated with numerous biological events such as inflammation, fibrosis, angiogenesis, and extracellular matrix deposition (4)(5). The fibrosis, with its feature of excessive deposition of extracellular matrix, has become an essential part of the pathology of leiomyomas, and is caused by various mechanisms including oxidative stress, hypoxia, cytokine, and growth factor activities (5). Moreover, estrogen can influence cancer pathogenesis and progression. It has been observed that estrogens affect the development of colorectal cancer (CRC) because of their interactions with estrogen receptors; ER- β , which is highly expressed in normal colonic tissue and becomes suppressed during tumor progression, plays an important role (3). Modulation of the expression of ER- β receptors by dietary phytoestrogens increases apoptosis and decreases tumorigenesis, indicating that phytoestrogens act as chemoprotective agents (3). Flavonoids, terpenoids, and polyphenols found in nature were shown to be capable of regulating cancer progression mechanisms including Wnt/ β -catenin, NF- κ B, PI3K/Akt/mTOR, and MAPK signaling pathways, thus preventing cell proliferation, angiogenesis, and metastasis in aggressive tumors, including TNBC (8).

Conventional medical therapies for endocrine system and other related disorders, such as HT and HRT, are common practices; however, they are commonly linked with several side effects such as cardiovascular disease, weight gain, dependency, increased cancer risk, and stroke (4,7)(9)(10). Likewise, conventional pharmaceutical therapies for disorders such as PMS, dysmenorrhea, UTIs, pregnancy-related problems, and lactation difficulties are also frequently associated with side effects and hence have resulted in the growing popularity of alternative medicine options (9)(11). The natural substances have traditionally been playing an important role in the healthcare systems all over the world and form the foundation of various traditional systems of medicine like Ayurvedic, Unani, Chinese, and other indigenous systems, where about 80 percent of the total world population uses herbal drugs for their healthcare purposes (6). For thousands of years, medicinal herbs were being used in traditional systems of medicine for the treatment of endocrinal and metabolic diseases. However,

despite the tremendous variety of plant types available, not more than a very few of them had ever been subjected to scientific investigation for their therapeutic potential (6).

Modern researchers thus found it necessary to study plants extensively and developed a new field called phytopharmacology, in which the natural active ingredients called phytochemicals found in plants were studied for their pharmacological actions. Phytochemicals include many different bioactive agents, like flavonoids, alkaloids, lignans, and polyphenols, that show a wide range of biological activities, such as antioxidation, anti-inflammation, antimicrobial action, neuroprotection, and hormonal regulation (4)(6). It is the effect on the hormone regulation that makes phytochemicals very significant for the proper functioning of the endocrine system (4).

Phytoestrogens are chemical substances derived from plants that share similarities in structure with endogenous estrogens. They can interact with estrogen receptors, producing either estrogenic or anti-estrogenic effects based on their specific tissue locations and distributions (3). These substances, which include isoflavones, lignans, and coumestans, have been linked to positive effects such as reducing menopause-related symptoms, promoting bone health, improving cognitive performance, and decreasing the risks of hormone-dependent cancers, specifically breast cancer and endometrial cancer (10). Notably, phytoestrogens can bind to estrogen receptors and control estrogenic activity (21). Examples of common phytoestrogens are isoflavones (genistein and daidzein), which are present in soy products and can affect hormone activity and reproduction (21).

Other effects of phytochemicals include their impact on the biosynthesis of hormones, particularly steroid hormones. Specific types of flavonoids can decrease steroid biosynthesis and hormone levels (21). Moreover, phytoestrogens selectively modulate estrogen receptors, which enables their beneficial action on certain tissues, such as the brain and bones, and reduced proliferative activity in estrogen-sensitive organs such as the uterus (2). The ability of phytochemicals to both interact with receptors and enzymes makes them very significant for endocrine modulation.

Finally, hormone regulation is strongly connected to precursor compounds, one of which is dehydroepiandrosterone (DHEA). It acts as a substrate from which most significant hormones such as testosterone and estradiol are synthesized. The studies showed that the administration of DHEA results in a significant increase in the concentration of the aforementioned hormones, especially at higher doses (18).

Phytochemicals can also modulate the work of the central nervous system due to their ability to improve mood. Antidepressant and anxiolytic activity of numerous medicinal plants has been demonstrated through their impact on neurotransmitter systems, neuroplasticity regulation, and modulation of HPA axis activity (7)(19). For example, *Crocus sativus*, *Lavandula angustifolia*, and *Curcuma longa* were found to influence serotonin synthesis and neuroprotection (19).

Hormonal imbalance due to Androgen is also considered another major factor for endocrine dysfunction, especially in females. Hormonal disorders such as PCOS, hirsutism, and acne are linked to excessive amounts of androgens (20). Anti-androgens inhibit the functions of androgens like

testosterone by interfering with their actions on different axes within the endocrine system (20). Various plant-based anti-androgens, including spearmint, green tea, licorice, and saw palmetto, have proven capabilities of regulating the activities of androgens through enzyme inhibition or conversion reactions (20).

Certain medicinal plants exhibit potent abilities to modulate the endocrine system. One of the most commonly used plants in traditional medicines to enhance reproductive health and hormonal regulation is *Tribulus terrestris* (15). Likewise, *Withania somnifera* (Ashwagandha) works as an adaptogen to relieve stress, improve hormone production, including DHEA and testosterone (17). Adaptogens assist in bringing physiological stability by improving resistance to stress and endocrine system function (13). Moreover, some plants, such as *Persicaria hydropiper*, possess biologically active substances including flavonoids, alkaloids, and steroids that have been shown to be effective in promoting estrogenic, antifertility, and healing activities under different traditional and experimental conditions (6).

The natural compounds also act on steroid hormone receptors by serving as weak agonists and antagonists; hence, they have an impact on hormone-dependent pathways involved in cancer and other diseases (7). Some flavonoids, including apigenin and naringenin, exert weak progestational effects, while others exert antiandrogenic and antiproliferation actions, thus enabling them to play a significant role in cancer prophylaxis and treatment (7).

Although phytochemicals show promising pharmaceutical potential, existing approaches for drug discovery are associated with high costs, lengthy experiments, and insufficient scalability (15). In addition, due to numerous phytochemicals and the great variability of their chemical structure, the isolation of active substances poses additional challenges (15). Consequently, advanced tools are needed for analyzing and prioritizing promising compounds.

Phytopharmacology is an innovative approach to drug discovery that makes use of bioinformatics, cheminformatics, and machine learning algorithms for large-scale analysis of phytochemical data sets, predicting interactions of compounds with target receptors, identifying active molecules, and improving the drug development process overall.

The increasing number of endocrine-related diseases and deficiencies in existing treatment options have made it clear that it is necessary to find novel approaches to dealing with the situation by combining traditional medicine with conventional healthcare practices (12). Nevertheless, even though many plant-based products are widely used and considered safe, there are serious concerns about their effectiveness, side effects, drug interaction profiles, and pharmacodynamics that require further investigation (9)(11). In this current research, phytochemicals have been collected from numerous scientific articles, and a systematic filtering was conducted considering their relevance in providing support to premenopausal endocrine function. This particular screening included phytochemicals having potential effects related to regulation of hormones, antioxidants, and endocrine and

neuroendocrine pathways. Afterward, through machine learning methods, these phytochemicals have been analyzed to screen out those that could be used in clinical settings as therapeutic agents.

Therefore, this research is a blend of traditional knowledge regarding medicinal plants and modern scientific techniques to further explore the possibility of finding more plant-based drugs and therapies. The objective of conducting this research is to discover and develop new plant-based treatment methods that can help in achieving better health care for patients suffering from health disorders due to hormonal imbalance or cancers. The combination of traditional knowledge with modern research has clearly highlighted the great potential of phytoestrogens and phytochemicals in treating women-related diseases and hormone disorders and even cancers in general (2)(3)(4)(5)(6)(7)(8)(9)(10)(11)(12).

Chapter-2

Aim and Objectives

A systematic strategy for the identification, screening, and prioritization of bioactive phytochemicals that can be used for the modulation of endocrine pathways for pre-menopause support utilizing computational phytopharmacology, machine learning, and molecular docking techniques.

1. Gathering and curating data on the phytochemicals from various resources pertaining to the health of the endocrine system.
2. Retrieving and standardizing the structure of the compounds by converting it into SMILES notation.
3. Prediction and evaluation of chemical-protein interactions within endocrine pathways using bioinformatics techniques.
4. Prediction of the toxicity profile of the selected compounds.
5. Pharmacokinetics analysis of selected compounds using ADME prediction.
6. Classification and prioritization of selected compounds using unsupervised machine learning technique using molecular descriptors.
7. Construction and analyzing protein-protein interaction networks.
8. Molecular docking studies of prioritized phytochemicals with key targets proteins.

Chapter 3

Review of Literature

Premenopausal women commonly experience hormonal imbalances characterized by irregular menstrual cycles, mood fluctuations, and metabolic disturbances. Conventional therapeutic approaches such as hormone replacement therapy (HRT) are widely used to manage these conditions; however, their long-term use has been associated with several adverse effects, including increased risk of cardiovascular diseases and hormone-dependent cancers. These limitations have driven the search for safer and more effective alternatives.

Phytochemicals, which are naturally occurring bioactive compounds derived from plants, have gained significant attention due to their ability to modulate endocrine pathways and exert multi-target therapeutic effects. Several studies have reported that phytochemicals such as flavonoids, alkaloids, and phenolic compounds can interact with hormone receptors, influence enzyme activity, and regulate signaling pathways involved in endocrine function. Their relatively low toxicity and broad biological activity make them promising candidates for managing premenopausal symptoms.

With the advancement of computational biology, *in silico* approaches have emerged as powerful tools in drug discovery and phytopharmacology. Databases such as chemical repositories enable the collection and standardization of phytochemical data using molecular representations like SMILES, facilitating large-scale computational analysis. Predictive tools further allow the identification of potential chemical–protein interactions, providing insights into the biological relevance of compounds.

Toxicity prediction and pharmacokinetic evaluation (ADME analysis) are essential steps in early-stage drug discovery to ensure safety and efficacy. Computational tools enable rapid screening of large compound libraries, helping to eliminate molecules with unfavorable toxicity profiles and poor bioavailability, thereby improving the efficiency of candidate selection.

Machine learning techniques have recently been integrated into phytochemical research to enhance compound prioritization. Unsupervised learning methods, such as K-means clustering, allow the classification of compounds based on molecular descriptors, enabling the identification of structurally and functionally similar groups. This data-driven approach supports the selection of promising candidates for further analysis.

In addition, protein–protein interaction (PPI) network analysis provides a systems-level understanding of biological processes. By constructing interaction networks, key proteins involved in endocrine regulation can be identified, along with their functional associations. Network analysis tools facilitate the detection of hub proteins and critical pathways, which are essential for understanding disease mechanisms and therapeutic targeting.

Finally, molecular docking has become an important computational technique to evaluate the interaction between bioactive compounds and target proteins at the molecular level. Docking studies help predict binding affinity, interaction stability, and the potential mechanism of action of selected compounds. This step serves as a crucial validation stage in computational drug discovery, supporting the identification of lead candidates for further experimental studies.

Overall, the integration of phytochemistry, bioinformatics, machine learning, and molecular docking provides a comprehensive framework for identifying and prioritizing potential therapeutic agents for premenopausal endocrine support.

Chapter 4

Materials and Methods

4.1 Experiment Design and Computational Pipeline

In this study, a computational phytopharmacology approach was applied to predict the possible plant-based bioactive compounds that could be employed to support the endocrine system of pre-menopausal females. The complexity and multifactorial nature of endocrine system regulation required designing a series of successive in silico phytochemical selection methods according to their chemical structure and biological activity, including safety, physicochemical properties, and biological importance.

The computational pipeline comprised several successive steps, which included phytochemical extraction from the literature resources, chemical structure acquisition, predicting chemical-protein interactions, toxicity evaluation, pharmacokinetic profiling, clustering based on molecule descriptors and analysis of protein-protein interaction networks. Thus, each step of the pipeline was aimed at gradually refining the dataset of phytochemicals to retain only safe and biologically meaningful compounds. The described computational pipeline was helpful in ranking phytochemicals before their subsequent docking.

4.2 Collection and Curation of Phytochemical Dataset

The first step in the computational workflow was the development of a dataset of phytochemicals that have been reported to affect endocrine regulation and hormone regulation. This was done through conducting an extensive literature search to find bioactive plant compounds related to reproductive health, menstrual regulation, metabolic balance and hormone signaling pathways.

A scientific literature search using terms such as phytochemicals, phytoestrogens, plant metabolites, endocrine modulation, hormonal regulation, and menstrual health among others was conducted in scientific databases such as PubMed, Google Scholar, Science Direct and Springer Link. Literature sources including peer reviewed papers and phytochemical databases were studied to identify phytochemicals that may have potential endocrinerelated effects.

As a result of this literature search, a total of 349 phytochemicals were identified. In order to develop a highquality dataset without duplication, the phytochemicals identified were manually curated. This was achieved by identifying duplicate phytochemicals based on their common name and chemical identifiers. All phytochemicals with ambiguous chemical identity were eliminated from the list.

4.3 Retrieval of Chemical Structure and SMILES Representation

Chemical structure representation is crucial in computational evaluation of small molecules. The Simplified Molecular Input Line Entry System (SMILES) was utilized to generate the representation of the chemical structures of the curated phytochemicals. Canonical SMILES codes are necessary in

cheminformatics analysis, toxicological assessment, pharmacokinetics modeling, and machine learning-based screening.

Canonical SMILES codes of individual phytochemicals were extracted from the ChEMBL database, which is a manually curated resource containing biologically active molecules and drug-like compounds. Every compound in the list of curated phytochemicals was individually searched in the database, and its corresponding canonical SMILES code was extracted.

Phytochemicals without SMILES information were not included in the analysis because structural representation is necessary in all the computational methods involved in the present research work. After the exclusion process, 188 phytochemicals were identified with successful retrieval of canonical SMILES codes.

4.4 Prediction of Chemical–Protein Interactions

In order to establish biological relevancy for the phytochemicals, a chemical-protein interaction analysis was carried out using the STITCH database. In STITCH, interactions are predicted based on experimental findings, curating pathways, literature mining, and modeling.

Phytochemicals with validated SMILES were used in the analysis using names or structures through the STITCH database. The organism of interest was set to be *Homo sapiens*. An interaction network was created with medium to high confidence score predictions to increase accuracy. Those phytochemicals without any predicted interactions with human proteins were not used for further study due to lack of biological importance with regards to human endocrine modulation.

This resulted in a total of 170 phytochemicals having predicted interactions with at least one human protein.

4.5 In Silico Toxicity Prediction

Predicting toxicity early is one of the vital steps required in computational drug discovery, especially when designing compounds for long-term consumption for hormonal support. The shortlisted phytochemicals were therefore screened for toxicity using the ProTox-II web server. The ProTox-II tool is capable of predicting different toxicological endpoints from molecular structure using machine learning methods.

The SMILES strings of the 170 phytochemicals were uploaded onto the ProTox-II web portal for predictions of acute oral toxicity and other toxicological endpoints like hepatotoxicity, carcinogenicity, immunotoxicity, mutagenicity and cytotoxicity. The key filtering parameter utilized here was toxicity class that categorizes compounds based on their LD₅₀ estimates in six toxicity classes.

The definition of these toxicity classes include:

- Class 1 – Fatal if swallowed
- Class 2 – Fatal in low doses
- Class 3 – Toxic if swallowed
- Class 4 – Harmful if swallowed
- Class 5 – May be harmful if swallowed

- Class 6 – Non-toxic

In the current case, any compound with toxicity classes 4, 5 or 6 were selected since this implies relative toxicity classes. Toxicity classes above these were not included in the list. After toxicity screening, 140 phytochemicals met the toxicity criteria.

4.6 ADME and Drug-Likeness Analysis

After toxicity testing, the selected phytochemicals underwent pharmacokinetic profiling to determine whether they had the potential to be considered viable treatment options. Predictions regarding the Absorption, Distribution, Metabolism, Elimination, and Toxicity (ADMET) characteristics of the shortlisted phytochemicals were conducted via SwissADME, an online platform frequently employed to predict the pharmacokinetics and drug-likeness of compounds.

The SMILES of the 140 non-toxic phytochemicals were subjected to ADMET predictions in the SwissADME server. Various pharmacokinetic assessments were carried out, such as gut absorption, lipophilicity, blood-brain barrier penetration, hydrophilicity, and interactions with the cytochrome P450 enzymes. The pharmacokinetic prediction was complemented by drug-likeness prediction based on established principles of medicinal chemistry.

Drug-likeness evaluation was based on the following criteria:

Lipinski's Rule of Five for oral drug-likeness prediction based on

- Molecular weight ≤ 500 Da
- $\text{LogP} \leq 5$
- Hydrogen bond donors ≤ 5
- Hydrogen bond acceptors ≤ 10

Veber's Rule, based on pharmacokinetic evaluation,

- Number of rotatable bonds ≤ 10
- Topological polar surface area $\leq 140 \text{ \AA}^2$

Ghose and Egan filters, which determine drug-likeness and permeability properties.

Those compounds that displayed undesirable pharmacokinetic profile or had more than one violation in drug-likeness criteria were filtered out. Upon ADME filtering, a total of 139 phytochemicals exhibiting desirable pharmacokinetic and drug-likeness profile were selected for molecular descriptors-based clustering.

4.7 Molecular Descriptor Extraction

To enable machine-learning-based classification of phytochemicals, key physicochemical descriptors were extracted from the SwissADME output. Molecular descriptors provide quantitative information about chemical structure and are widely used in cheminformatics to study relationships between molecular properties and biological activity.

For the present study, four fundamental descriptors were selected based on their relevance to drug discovery and pharmacokinetics:

- Molecular weight
- Lipophilicity (LogP)
- Hydrogen bond donors (HBD)
- Hydrogen bond acceptors (HBA)

These descriptors influence membrane permeability, solubility, oral bioavailability and molecular interactions with biological targets. The descriptor dataset obtained for the 139 phytochemicals was prepared for machine learning analysis.

4.8 Machine Learning-Based Clustering Using KNIME

Unsupervised machine learning analysis was performed using the KNIME Analytics Platform to classify phytochemicals based on similarity in physicochemical properties. Prior to clustering, the descriptor dataset underwent preprocessing to ensure data quality and comparability.

The dataset was normalised and standardised to remove scale bias and ensure that each descriptor contributed equally to the clustering process. Missing values were checked and corrected to maintain dataset integrity.

K-means clustering was then applied to group the phytochemicals into clusters based on similarity in molecular descriptors. The number of clusters (k) was set to three, and Euclidean distance was used as the similarity metric. This approach enabled the grouping of compounds with comparable physicochemical profiles, facilitating the identification of structurally and pharmacologically similar phytochemicals.

To visualise clustering behaviour, parallel coordinates plots were generated to illustrate variation across molecular descriptors, and scatter plots were created to visualise clustering patterns based on molecular weight and lipophilicity.

4.9 Construction of Protein–Protein Interaction Network

To understand the biological relationships among the predicted protein targets, a protein–protein interaction (PPI) network was constructed. Analysing interactions among target proteins helps identify shared pathways, functional associations and key regulators involved in endocrine-related biological processes.

Protein targets obtained from the chemical–protein interaction analysis were submitted to the STRING database for PPI network construction. The organism was restricted to *Homo sapiens* to ensure relevance to human physiology. Interaction networks were generated using medium to high confidence scores to improve reliability of predicted associations.

STRING integrates multiple sources of interaction evidence, including experimental data, co-expression analysis, gene co-occurrence, gene fusion events and literature mining. The resulting network consisted of nodes representing proteins and edges representing functional associations

between them. The generated interaction network was exported for further visualisation and topological analysis.

4.10 Network Analysis Using Cytoscape

The PPI network generated from STRING was imported into Cytoscape software for advanced visualisation and network analysis. Cytoscape enables topological analysis of biological networks and identification of important nodes and clusters within complex interaction maps.

4.10.1 Module Detection Using MCODE

To identify densely connected regions within the network, the Molecular Complex Detection (MCODE) plugin was used. MCODE detects highly interconnected clusters that often represent functional biological modules or protein complexes. The identification of such modules helps reveal groups of proteins involved in related biological processes.

4.10.2 Hub Gene Identification Using CytoHubba

To determine the most important proteins within the interaction network, hub gene analysis was performed using the CytoHubba plugin. Network centrality measures were used to rank proteins based on their topological importance.

Four centrality algorithms were applied:

- Maximal Clique Centrality (MCC)
- Degree centrality
- Closeness centrality
- Betweenness centrality

The top ten genes identified by each method were compared, and overlapping genes across all four algorithms were determined using a Venn diagram approach. Genes common to all methods were considered highconfidence hub genes and were selected for further downstream analysis.

4.11. Molecular Docking Analysis

Molecular docking was performed to evaluate the binding affinity of selected phytochemicals against the identified hub proteins (**TP53, TNF, IL6, and CTNNB1**). The docking workflow involved ligand preparation, protein preparation, docking simulation, and interaction analysis.

4.11.1 Ligand Preparation

The selected phytochemicals were retrieved from the **PubChem** database.

- Ligands were downloaded in **3D structure format (SDF)**
- Structural integrity of each compound was verified
- Ligands were prepared for docking by ensuring proper geometry and format compatibility

These prepared ligands were used as input for molecular docking analysis.

4.11.2 Protein Structure Retrieval

The three-dimensional structures of the target proteins (**TP53, TNF, IL6, and CTNNB1**) were obtained from the **RCSB Protein Data Bank**.

Selection of protein structures was based on:

- Homo sapiens origin
- High structural resolution
- Structural completeness

The downloaded protein structures were used for further preparation and docking studies.

4.11.3 Protein Preparation

Protein preparation was carried out using **Discovery Studio Visualizer**.

The following preprocessing steps were performed:

- Removal of water molecules
- Removal of co-crystallized ligands (if present)
- Addition of hydrogen atoms
- Correction of structural inconsistencies

These steps ensured that the protein structures were in a suitable form for docking analysis.

4.11.4 Molecular Docking Procedure

Molecular docking was performed using **PyRx**, which integrates AutoDock Vina for docking simulations.

- Prepared protein and ligand structures were imported into PyRx
- Ligands were converted into appropriate docking format
- Docking simulations were performed for each ligand–protein pair
- Binding affinity values were obtained in terms of **binding energy (kcal/mol)**

Lower (more negative) binding energy values indicate stronger ligand–protein interactions.

4.11.5 Interaction Analysis and Visualization

The docking results were analyzed and visualized using **Discovery Studio Visualizer**.

- **2D interaction diagrams** were generated to identify interacting amino acid residues
- **3D visualization** was used to observe ligand orientation within the binding pocket
- Types of interactions such as:

- hydrogen bonding
 - hydrophobic interactions
 - π -alkyl interactions
- were examined

This analysis provided insights into the binding mode and interaction stability of the ligand–protein complexes.

4.11.6 Selection of Top Docking Hits

For each target protein, the top-performing phytochemicals were selected based on:

- lowest binding energy values
- interaction stability
- consistency of binding across targets

The top two compounds for each protein were shortlisted for further analysis.

Chapter 5

Results

5.1. Data Retrieval (Phytochemical Dataset Collection)

Phytochemicals were systematically collected from **published scientific literature**, including research articles, review papers, and phytochemical databases. The aim of this step was to construct a **comprehensive and diverse dataset** of plant-derived bioactive compounds with potential relevance to endocrine modulation.

A total of **349 phytochemicals** were compiled during this stage. The selection was based on reported biological activity, traditional medicinal usage, and relevance to hormonal or metabolic regulation.

This step is crucial because:

- It ensures **broad chemical diversity**, increasing the likelihood of identifying effective candidates
- It provides a **literature-backed dataset**, improving the scientific validity of the study
- It forms the foundation for all subsequent computational analyses

However, datasets obtained from literature often contain redundancy and inconsistencies, which necessitates further preprocessing.

5.2. Duplicate Removal and Data Cleaning

The initial dataset was subjected to **data preprocessing and cleaning** to remove duplicate entries. Duplicate compounds commonly arise when the same phytochemical is reported across multiple studies or sources.

Duplicates were identified based on:

- compound names
- structural similarity
- identical chemical records

After removing duplicate entries, the dataset was refined to include only **unique phytochemicals**, ensuring that each compound was represented only once.

This step is essential because:

- It prevents **data redundancy**, which can bias computational analysis
- It ensures **accuracy in clustering and statistical evaluation**

- It improves **computational efficiency** by reducing unnecessary repetitions

The removal of duplicates enhances the reliability of downstream analyses, particularly machine learning, where repeated data points can distort clustering results.

5.3. SMILES Retrieval and Validation

Following data cleaning, structural information for each phytochemical was obtained in the form of **SMILES (Simplified Molecular Input Line Entry System)** from the **ChEMBL** database.

SMILES is a standardized, text-based representation of chemical structures that allows compounds to be processed computationally. Out of the curated dataset, only **188 phytochemicals were found to have valid SMILES representations**, and these were retained for further analysis.

The use of SMILES is critical because:

- It provides a **machine-readable format** of chemical structures
- It enables **molecular descriptor calculation**
- It serves as input for:
 - ADME analysis
 - toxicity prediction
 - machine learning algorithms

Compounds without valid SMILES were excluded, as they cannot be processed by computational tools and would compromise the integrity of the analysis.

This step ensures that the dataset is **computationally compatible and structurally defined**, which is essential for accurate and reproducible results.

5.4. Chemical–Protein Interaction Analysis (STITCH)

The filtered phytochemical dataset with valid structural information was subjected to **chemical–protein interaction analysis** using the **STITCH database**. STITCH integrates data from experimental studies, databases, and computational predictions to identify interactions between chemicals and proteins.

Out of the 188 phytochemicals analyzed, **170 compounds were predicted to interact with biological targets**, indicating a high level of potential biological relevance.

This step plays a critical role in the workflow as it:

- Establishes a link between **chemical structures and biological targets**
- Ensures that selected compounds are not merely structurally valid but also **functionally significant**
- Helps identify proteins involved in **endocrine signaling and regulatory pathways**

The high proportion of interacting compounds suggests that the curated phytochemicals possess **substantial biological activity**, making them suitable candidates for further screening.

5.5. Toxicity Prediction (ProTox-II)

To ensure the safety of the selected phytochemicals, **in silico toxicity prediction** was performed using **ProTox-II**, a widely used tool for predicting various toxicity endpoints based on chemical structure.

From the 170 interacting compounds, **140 phytochemicals passed toxicity screening**, indicating acceptable safety profiles.

This step is particularly important because:

- Toxic compounds can lead to **failure in later stages of drug development**
- Endocrine-related therapies often require **long-term administration**, making safety a critical factor
- Early elimination of toxic compounds improves **efficiency and reliability** of the screening process

By filtering out potentially harmful compounds, this step ensures that only **safe and viable candidates** proceed to further analysis.

5.6. ADME and Drug-Likeness Analysis (SwissADME)

The pharmacokinetic properties of the phytochemicals were evaluated using **SwissADME**, which provides insights into drug-likeness and ADME (Absorption, Distribution, Metabolism, and Excretion) parameters.

Out of 140 compounds, **139 phytochemicals demonstrated acceptable ADME properties**, indicating favorable pharmacokinetic profiles.

Key aspects evaluated include:

- Gastrointestinal absorption
- Lipophilicity (LogP)

- Hydrogen bonding characteristics
- Bioavailability

This step ensures that selected compounds are:

- **Absorbable and bioavailable**
- Suitable for **systemic circulation**
- Likely to exhibit **effective therapeutic activity**

The high retention rate at this stage suggests that most phytochemicals possess **favorable drug-like characteristics**, supporting their potential as therapeutic agents.

5.7. Machine Learning Analysis (KNIME)

The refined dataset of 139 phytochemicals was subjected to **machine learning analysis using the KNIME Analytics Platform**.

An unsupervised learning approach, specifically **K-means clustering**, was applied to classify compounds based on molecular descriptors such as:

- Molecular weight
- LogP
- Hydrogen bond donors
- Hydrogen bond acceptors

The clustering process grouped phytochemicals into distinct clusters based on **similarity in physicochemical properties**.

This step is important because:

- It enables **data-driven classification of compounds**
- It identifies **patterns and relationships within the dataset**
- It helps prioritize compounds for further analysis

The use of unsupervised learning allows classification without prior labeling, making it suitable for exploratory analysis of complex biological datasets.

5.7.1 Visualization of Clustering Patterns

5.7.1.1 Parallel Coordinates Plot

The parallel coordinates plot (*fig 1*) demonstrated variation across molecular descriptors such as molecular weight, LogP, hydrogen bond donors, and acceptors.

Distinct patterns were observed across clusters, indicating that:

- each cluster possesses unique descriptor distributions
- compounds within a cluster share similar molecular characteristics

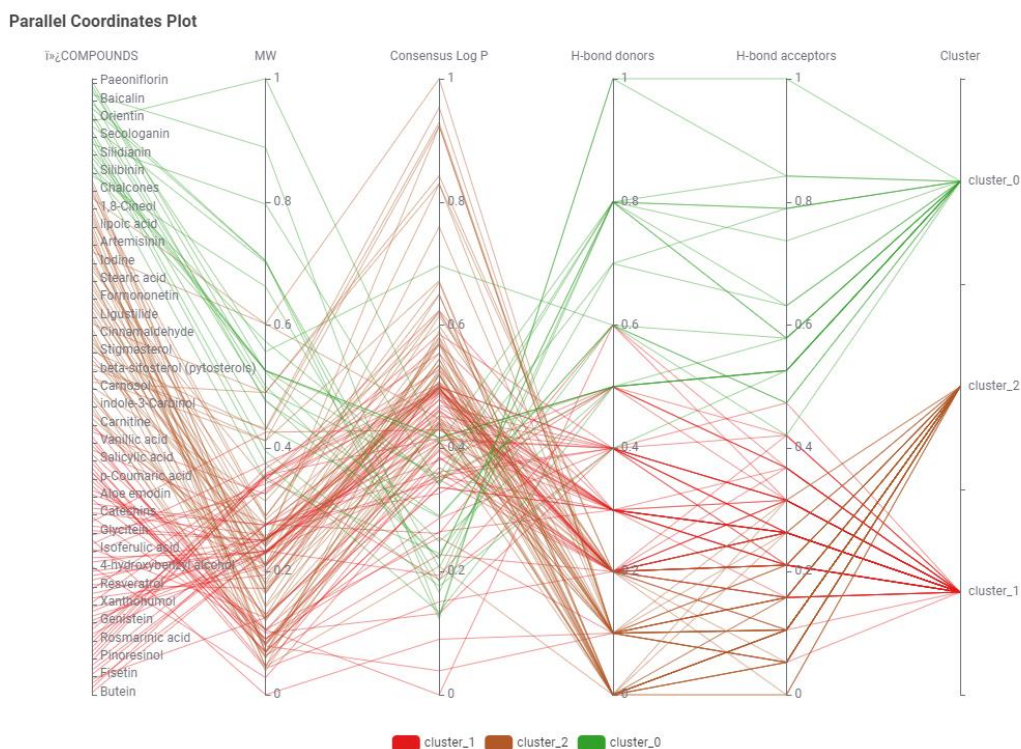


Figure 1: The parallel coordinates plot demonstrated variation across molecular descriptors such as molecular weight, LogP, hydrogen bond donors, and acceptors.

5.7.1.2 Scatter Plot Analysis

The scatter plot (*fig 2*) illustrated clustering based on molecular weight and lipophilicity (LogP).

The compounds were distributed into three clearly separated clusters, with Cluster-3 showing:

- tighter grouping
- reduced dispersion

This indicates that the clustering process effectively classified compounds based on key physicochemical parameters.

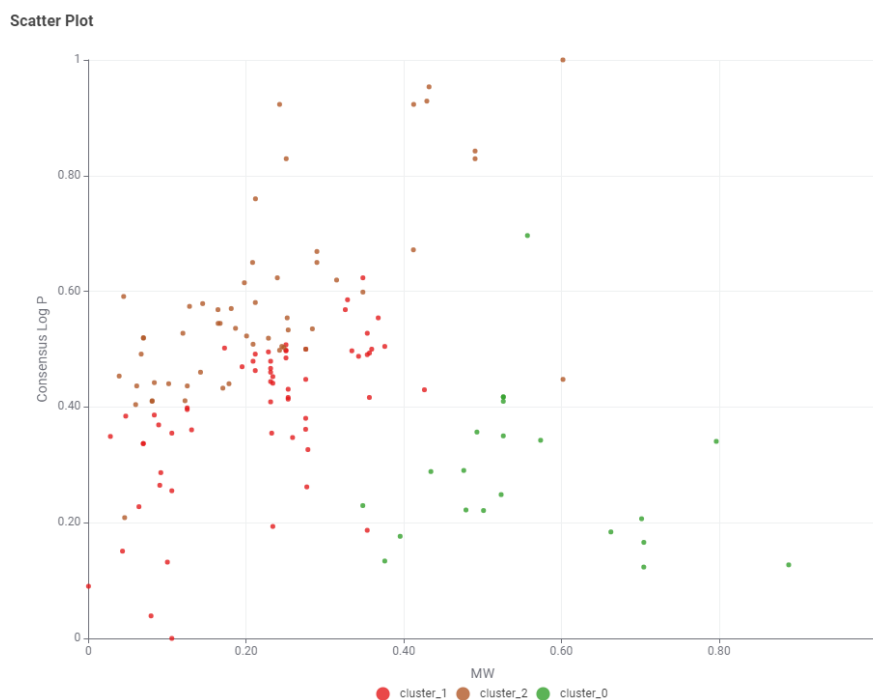


Figure 2, The scatter plot illustrated clustering based on molecular weight and lipophilicity (LogP).

5.8. Protein–Protein Interaction Analysis (STRING)

The protein targets identified from STITCH were further analyzed using STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) to construct a protein–protein interaction (PPI) network.

STRING integrates known and predicted protein interactions derived from:

- experimental data
- computational prediction
- text mining

The generated PPI network provides insights into:

- functional relationships among proteins
- pathway-level interactions
- biological processes associated with endocrine regulation

This step is critical for understanding how individual protein targets interact within a **larger biological system**, enabling a systems-level perspective of the study.

5.9. Network Analysis (Cytoscape)

The PPI network obtained from STRING was imported into **Cytoscape** for advanced visualization and analysis.

Two major analyses were performed:

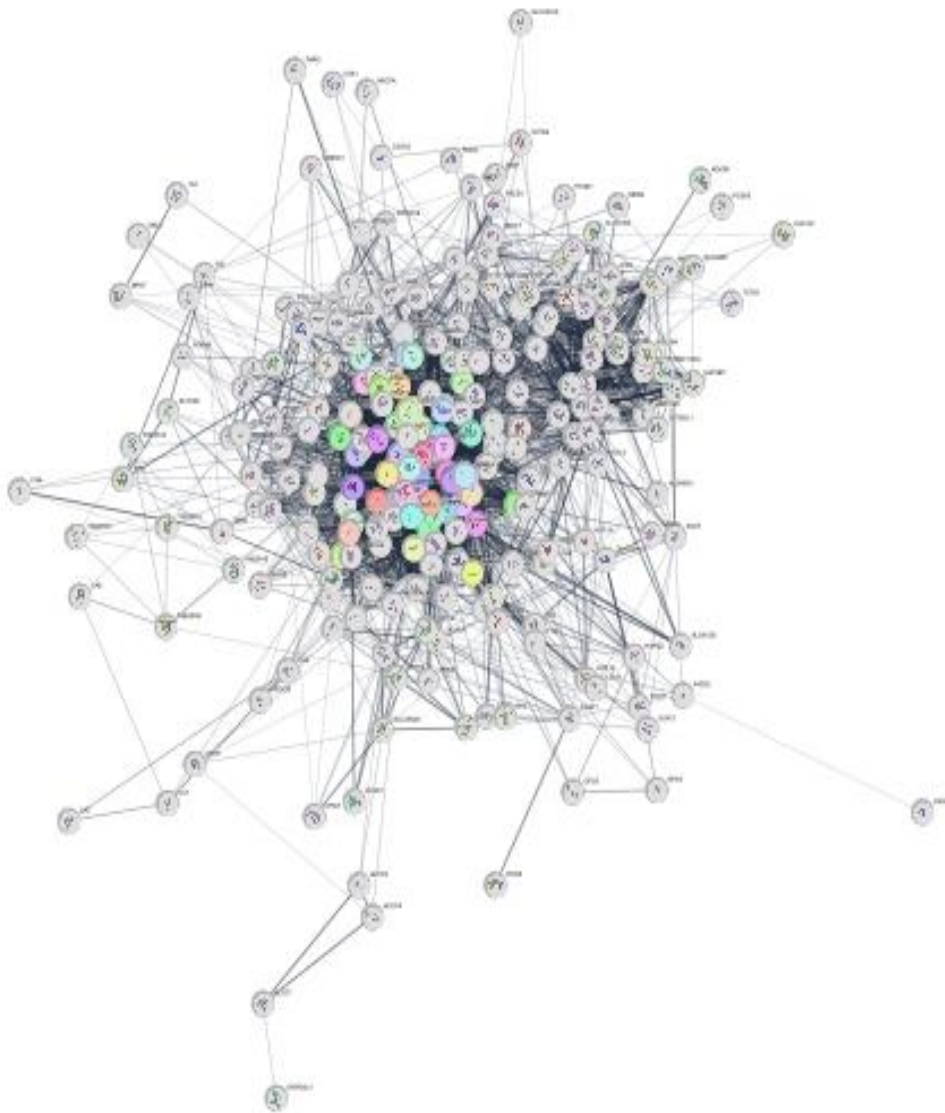


Figure 3 string network obtained from cytoscape

5.9.1 Module Detection (MCODE Analysis)

MCODE was used to identify **highly interconnected clusters (modules)** within the network. These modules represent groups of proteins that are functionally related and often participate in similar biological processes.

The identification of multiple clusters indicates:

- presence of **functional sub-networks**
- involvement of proteins in **coordinated biological activities**

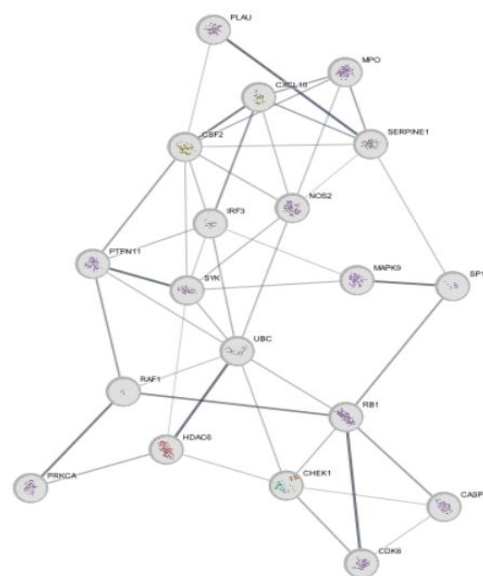


Figure 5. MCODE Cluster 2 identified from the PPI network (Score = 14.333; Nodes = 19; Edges = 129).

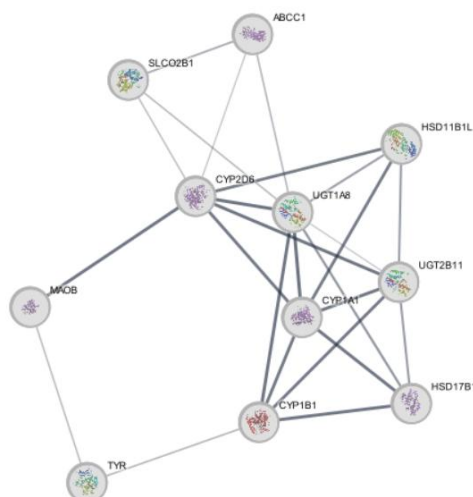


Figure 7. MCODE Cluster 4 identified from the PPI network (Score = 5.000; Nodes = 11; Edges = 25).



Figure 8. MCODE Cluster 5 identified from the PPI network (Score = 5.000; Nodes = 5; Edges = 10).



Figure 9. MCODE Cluster 6 identified from the PPI network (Score = 3.750; Nodes = 9; Edges = 15).

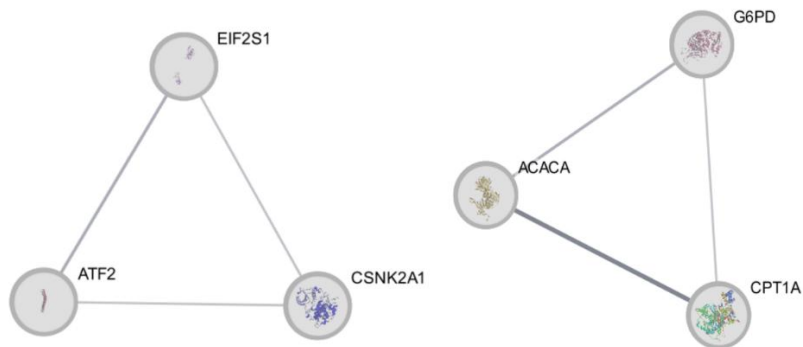


Figure 10. MCODE Cluster 7 identified from the PPI network (Score = 3.000; Nodes = 3; Edges = 3).

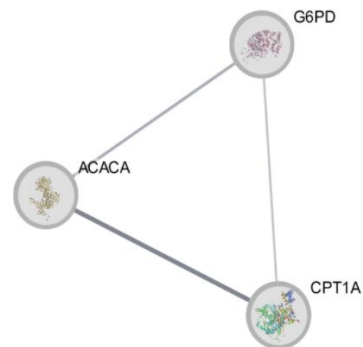


Figure 11. MCODE Cluster 8 identified from the PPI network (Score = 3.000; Nodes = 3; Edges = 3).

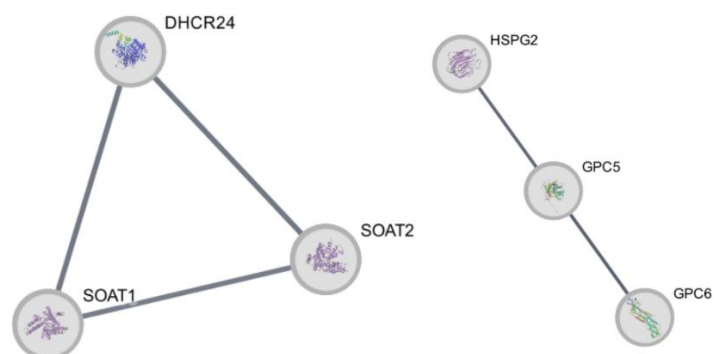


Figure 12. MCODE Cluster 9 identified from the PPI network (Score = 3.000; Nodes = 3; Edges = 3).

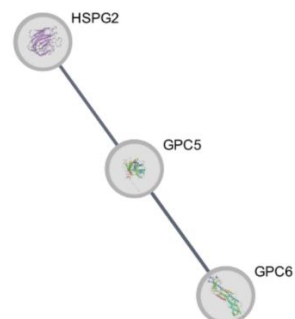


Figure 13. MCODE Cluster 10 identified from the PPI network (Score = 3.000; Nodes = 8; Edges = 28).

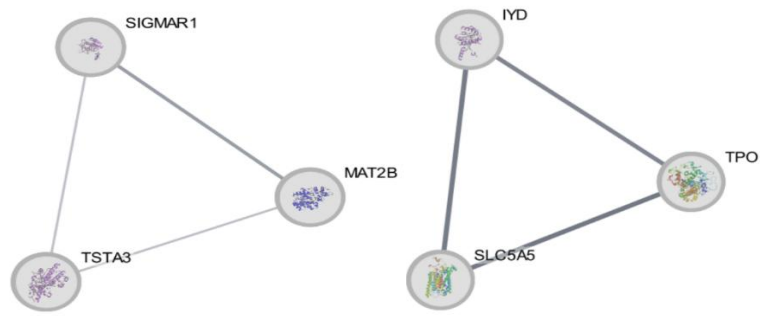
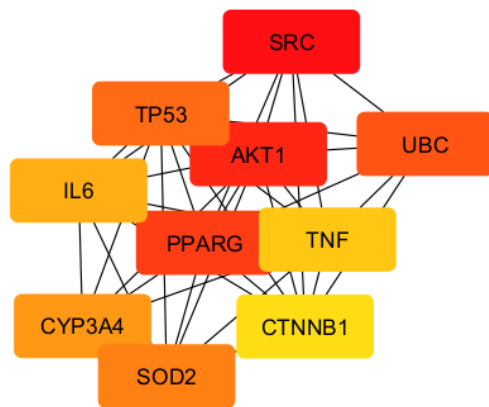


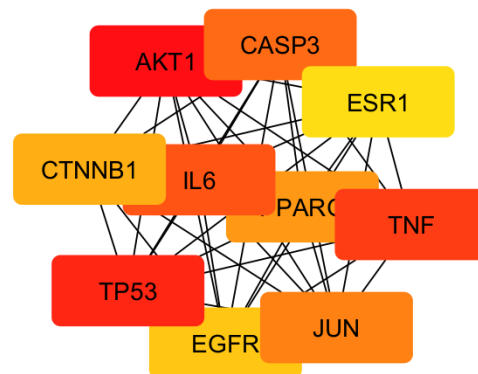
Figure 14. MCODE Cluster 11 identified from the PPI network (Score = 3.000; Nodes = 6; Edges = 3).

Figure 15. MCODE Cluster 12 identified from the PPI network (Score = 3.000; Nodes = 3; Edges = 3).

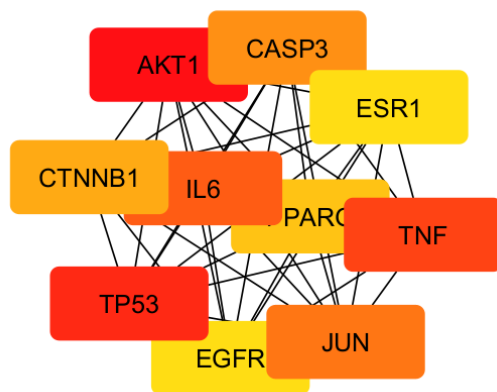
BETWEENNESS- TOP 10 HUB GENES



CLOSENESS- TOP 10 HUB GENES



DEGREE- TOP 10 HUB GENES



MCC- TOP 10 HUB GENES

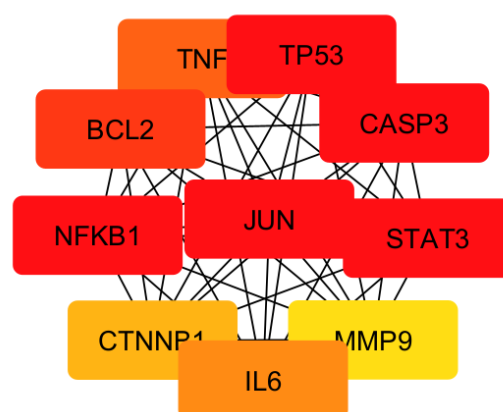


Figure 16: Top 10 hub genes identified using the Betweenness centrality method.

Figure 17: Top 10 hub genes identified using the Closeness centrality method.

Figure 18: Top 10 hub genes identified using the Degree method.

Figure 19: Top 10 hub genes identified using the MCC method.

5.9.2 Hub Gene Identification (CytoHubba Analysis)

CytoHubba was used to identify **hub genes** based on network centrality using multiple algorithms:

- MCC (Maximal Clique Centrality)
- Degree
- Closeness
- Betweenness

Hub genes are highly connected nodes that play a **central role in maintaining network structure and function**.

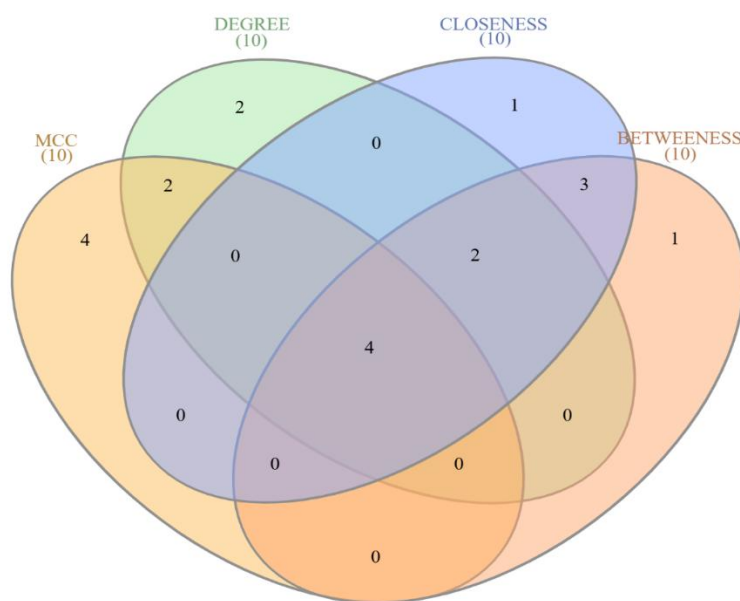


Figure 20: Venn diagram showing the overlap of top 10 hub genes identified by Degree, Closeness, Betweenness, and MCC algorithms.

To increase confidence, a **Venn diagram analysis** was performed to identify genes common across all methods. This resulted in the identification of four key hub genes:

TP53, TNF, IL6, CTNNB1

These genes represent **high-confidence regulatory targets** and are likely to play crucial roles in:

- endocrine signaling
- inflammatory pathways
- cellular regulation

5.10. Molecular Docking Analysis

Molecular docking was performed to evaluate the binding affinity of selected phytochemicals against the four hub proteins identified from network analysis, namely TP53, TNF, IL6, and CTNNB1. The docking study aimed to assess ligand–protein interactions by calculating binding energy values using PyRx software.

Ligand structures were obtained from PubChem, and protein structures were retrieved from the Protein Data Bank (PDB). Protein preparation and visualization were carried out using Discovery Studio Visualizer. The prepared ligands and proteins were subjected to docking, and binding affinities were obtained in terms of docking scores (kcal/mol).

The docking scores represent the interaction strength between ligand and protein, where more negative values correspond to stronger binding affinity.

TNF

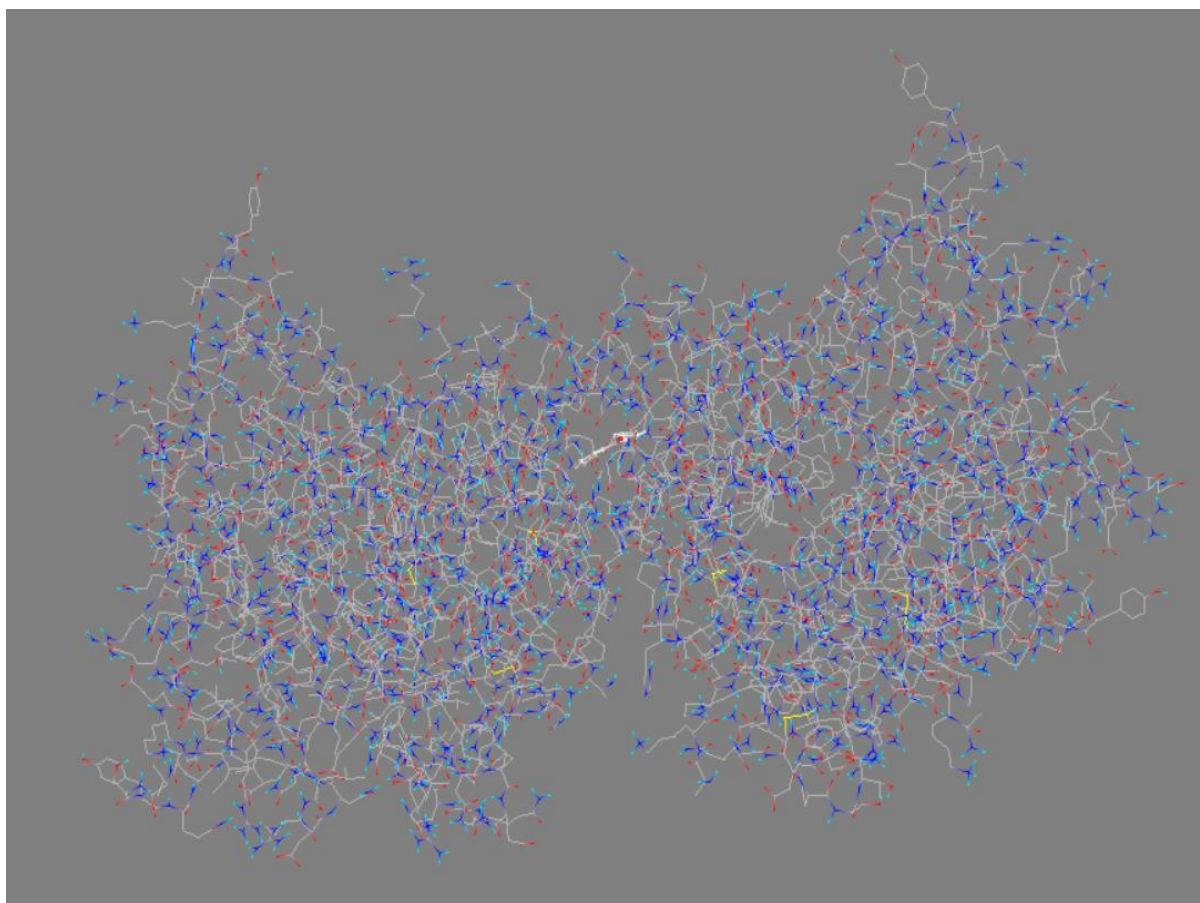


Figure 21: 3D visualization of the TNF–ligand complex showing ligand binding orientation; protein from PDB and docking performed using PyRx.

CTNNB1

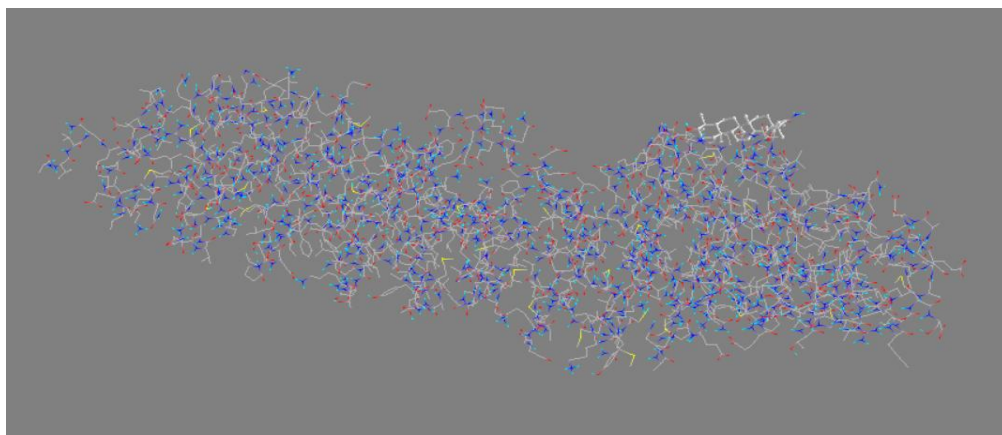


Figure 22: 3D visualization of the CTNNB1–ligand complex showing ligand binding orientation; protein from PDB and docking performed using PyRx.

TP53

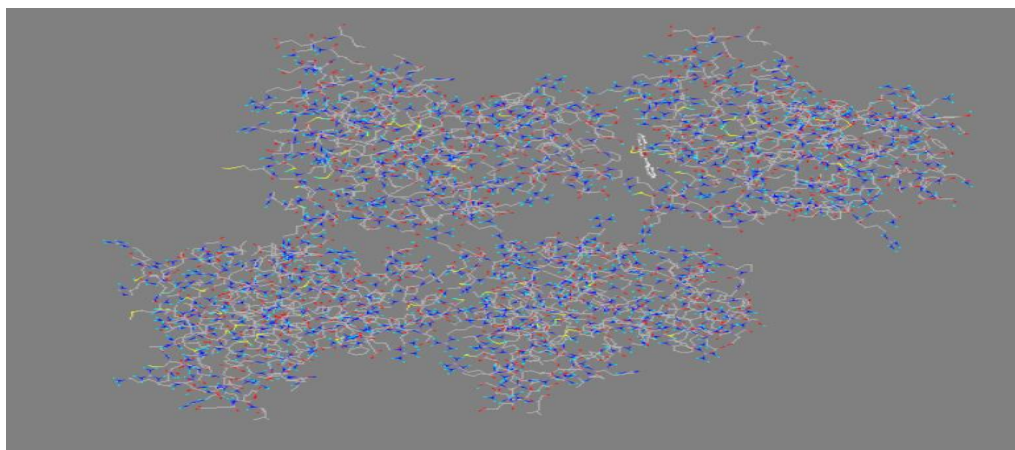


Figure 23: 3D visualization of the TP53–ligand complex showing ligand binding orientation; protein from PDB and docking performed using PyRx.

IL6

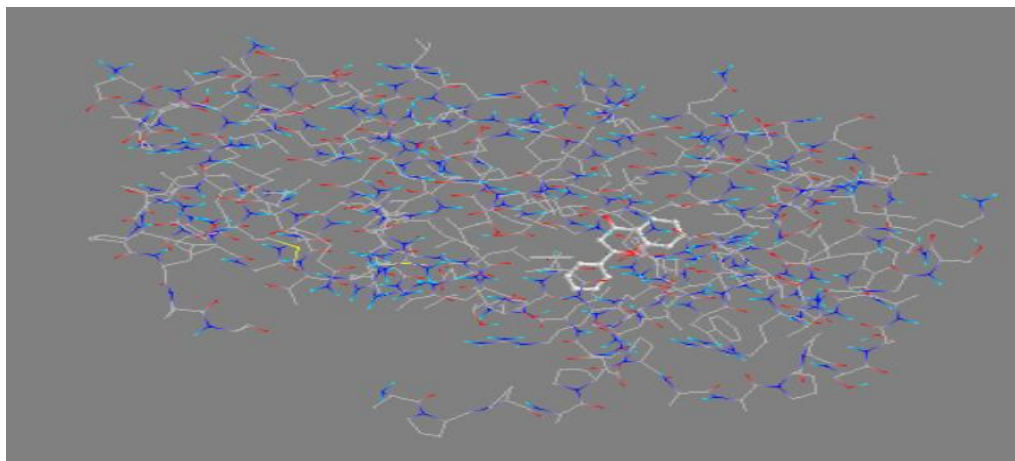


Figure 24: 3D visualization of the IL6–ligand complex showing ligand binding orientation; protein from PDB and docking performed using PyRx.

5.11. Docking Results for TNF

Docking analysis of phytochemicals against TNF resulted in variation in binding affinities among the tested compounds.

Among the docked ligands:

Campesterol exhibited the lowest binding energy of -12.3 kcal/mol

Artemisinin showed a binding energy of -10.8 kcal/mol

These values indicate that both compounds exhibited strong binding affinity toward TNF, with Campesterol showing the lowest docking score among the evaluated compounds.



Figure 25: 2D & 3D structure Campesterol exhibited the lowest binding energy of -12.3 kcal/mol

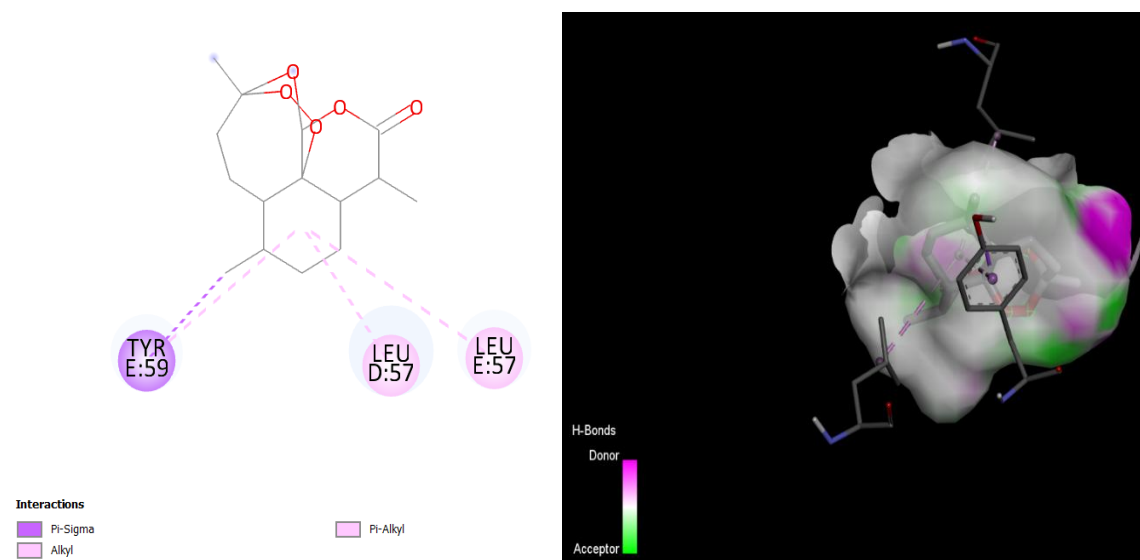


Figure 26: 2D & 3D structure of Artemisinin showed a binding energy of -10.8 kcal/mol

5.12. Docking Results for IL6

Docking analysis of phytochemicals against IL6 revealed the following binding affinities:

Ursolic acid showed a binding energy of -7.9 kcal/mol

Piperine showed a binding energy of -7.4 kcal/mol

Both compounds exhibited comparable docking scores, with Ursolic acid showing a lower binding energy compared to Piperine.

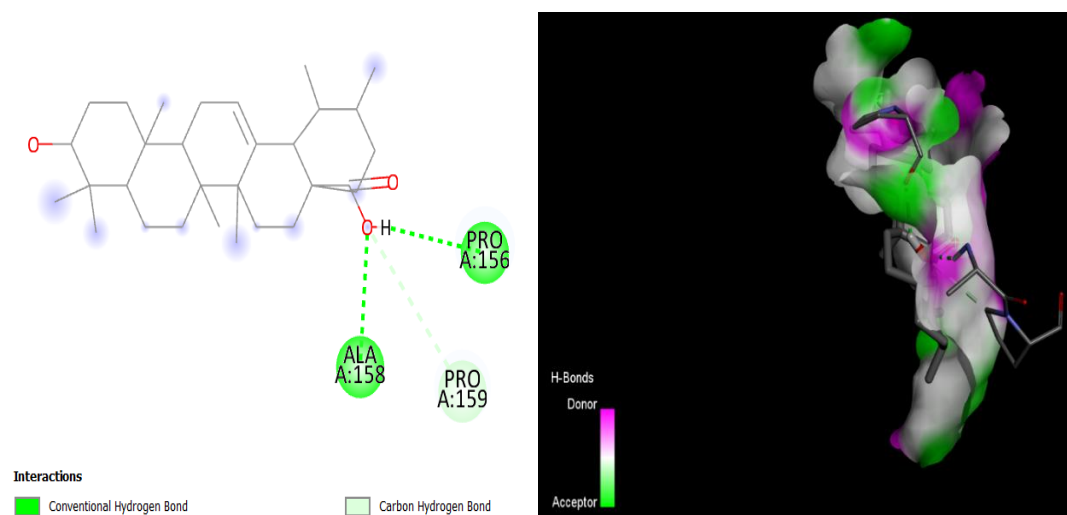


Figure 27: 2D & 3D of Ursolic acid showed a binding energy of -7.9 kcal/mol

5.13. Docking Results for TP53

Docking analysis of phytochemicals against TP53 showed:

Flavone exhibited a binding energy of -8.1 kcal/mol

Aurone exhibited a binding energy of -8.0 kcal/mol

The docking scores indicate that both compounds showed similar interaction values with minimal variation.

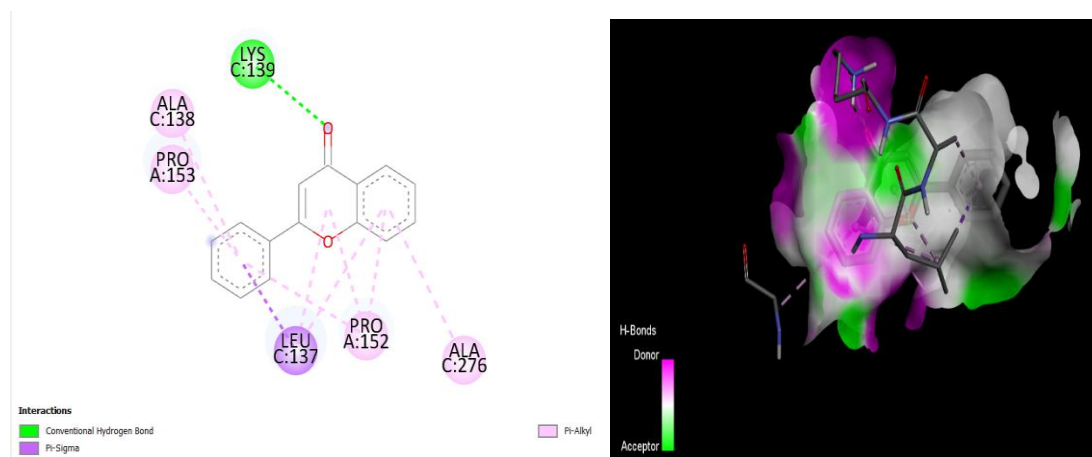


Figure 28: 2D & 3D tructure of Flavone exhibited a binding energy of -8.1 kcal/mol

5.14. Docking Results for CTNNB1

Docking analysis of phytochemicals against CTNNB1 revealed:

Ursolic acid exhibited a binding energy of -7.8 kcal/mol

Campesterol exhibited a binding energy of -7.2 kcal/mol

Ursolic acid showed a lower docking score compared to Campesterol.

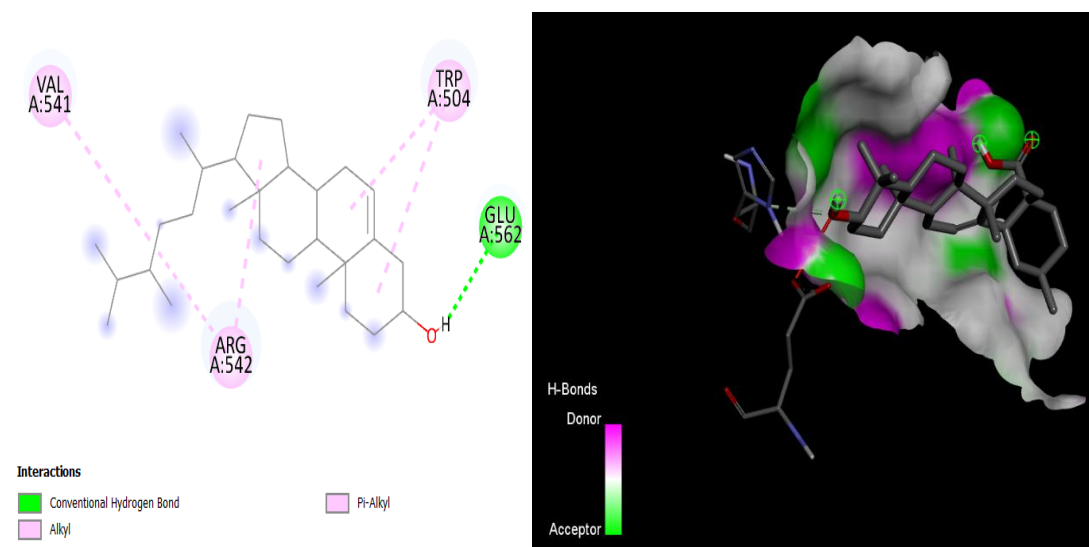


Figure 28: 2D & 3D Structure of Ursolic acid exhibited a binding energy of -7.8 kcal/mol

5.15. Comparative Docking Results Across Target Proteins

A comparison of docking scores across all four target proteins shows variation in binding affinity among different ligand–protein combinations.

The lowest binding energy observed in the study was -12.3 kcal/mol for TNF–Campesterol

TNF–Artemisinin also showed a strong binding energy of -10.8 kcal/mol

Docking scores for TP53 ranged from -8.0 to -8.1 kcal/mol

Docking scores for IL6 ranged from -7.4 to -7.9 kcal/mol

Docking scores for CTNNB1 ranged from -7.2 to -7.8 kcal/mol

The results indicate that binding affinities differ across target proteins and phytochemicals.

5.16. Selection of Top Docking Compounds

Based on docking scores, the top two phytochemicals for each protein were identified as follows:

TNF: Campesterol (−12.3 kcal/mol),

IL6: Ursolic acid (−7.9 kcal/mol), Piperine (−7.4 kcal/mol)

TP53: Flavone (−8.1 kcal/mol), Aurone (−8.0 kcal/mol)

CTNNB1: Ursolic acid (−7.8 kcal/mol), Campesterol (−7.2 kcal/mol)

These compounds were selected based on their docking scores among the screened phytochemicals.

Chapter 6

Discussion

For the current research, the integrated approach of computational phytopharmacology was used to discover and rank bioactive phytochemicals for potential use in supporting premenopausal endocrine system health. The multiple stages of the screening process allowed for the systematic decrease of the number of entries within the dataset while preserving its biological validity and safe use.

The initial set of phytochemicals containing 349 entries was gathered from scientific literature. Duplication removal contributed to the reliability of the data and avoided repetition of the same entry within the dataset. Afterward, the search for valid SMILES representation of molecules decreased the number of compounds to 188 entries, making it possible to work with chemical structures through machine learning and cheminformatics algorithms.

According to the results of protein-compound analysis by STITCH, the vast majority of the molecules (170 phytochemicals) showed predicted interaction with human proteins. These results indicate the substantial biological significance of the selected compounds in terms of pathways involved in endocrine regulation. In this way, multi-target activities can be considered a beneficial feature of phytochemicals that makes them appropriate for managing diseases involving complex systems.

As for the safety aspect of the selected compounds, 140 phytochemicals were recognized as potentially safe based on the results of toxicity prediction performed via ProTox-II. Safety assessment plays a critical role in selecting the compounds for endocrine therapy as this type of treatment implies prolonged intake with a minimal level of toxicity. High selection rate demonstrates the inherent safety characteristics of plant-derived substances compared to synthetic drugs.

According to the results of ADME and drug-likeness prediction via SwissADME, 139 compounds exhibited favorable pharmacokinetics. A majority of phytochemicals passed the drug-likeness test, complying with Lipinski's rules of five and Veber's parameters.

The use of machine learning algorithms in clustering using KNIME software successfully grouped the phytochemicals into three clusters according to molecular descriptors. It is clear from parallel coordinate and scatter plots that the clustering was done based on physicochemical properties. In particular, the formation of one tight group of compounds implies that the compounds have more significance pharmacologically.

Protein-protein interaction network analysis using STRING provided information about the target proteins at a systems level. The network was dense with many nodes, indicating functional association between the proteins participating in endocrine signaling and other biological functions. Analysis using Cytoscape software indicated that there were multiple functional modules in the network through MCODE clustering.

The hub genes were identified using CytoHubba software, which indicated TP53, TNF, IL6, and CTNNB1 as important regulatory genes. The reason for their selection is that they have a crucial role in regulating biological activities in cells, including inflammation, apoptosis, and Wnt/ β -catenin and NF- κ B signaling.

Molecular docking analysis offered more insights into the binding affinity of the chosen phytochemicals towards the hub proteins. From the docking experiments, Campesterol had the highest binding affinity towards TNF (-12.3 kcal/mol). Likewise, Artemisinin showed great binding towards TNF (-10.8 kcal/mol). Flavone and Aurone also displayed similar binding affinities with TP53. Ursolic acid had high binding affinity towards both IL6 (-7.9 kcal/mol) and CTNNB1 (-7.8 kcal/mol).

The difference in binding affinity with the other proteins shows the specificity of the ligand-protein interactions. Overall, from the docking results, it can be observed that the chosen phytochemicals have a great capability to interact with the key regulatory proteins within the endocrine pathways. Thus, the hypothesis that plant-based chemicals have therapeutic value in balancing hormonal levels is valid.

In summary, this investigation shows that the combination of computational approaches, machine learning algorithms, network analysis, and molecular docking is an effective strategy for prioritizing phytochemicals. Nonetheless, *in silico* techniques should be followed by *in vivo* experiments to validate the *in silico* results.

Chapter 7

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